Adilet Shynggys

Mechanical & Aerospace Engineering Student

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Skills

- **Fabrication:** 3D printing, electrospinning, silver-ink & silicon printing
- **Software:** SolidWorks (CSWA), MATLAB, Python, OriginPro, Mathematica
- **Technical:** Thermodynamics, Fluid Mechanics, Structural Mechanics, Energy Systems, Nuclear Power Plant Design
- **Languages:** English (Fluent), Russian (Fluent), Kazakh (Native)

Awards & Honors

- Presidential Scholarship, Republic of Kazakhstan (2026)
- Scholarship of Nazarbayev University's President
- Mereke CF Scholarship
- Yessenov Scholarship (Top 20 of 500+ applicants)
- Best Research Poster — B&R Initiative Conference (2025)
- 1st Place — Mechaton Engineering Case Competition
- Champion — NU AIChE Engineering Challenges (2 events)
- Dean's List (4 consecutive semesters)

Leadership & Volunteering

- **Charity Trail Run Organizer:** Raised 600K KZT (∼1,300 USD) with 150+ participants.
- **Physics Tutor & Mentor:** Guided 20+ freshman students, boosting average grades by 15%.
- **Volunteer:** HKU Muslim Society Ramadan events; Mereke CF "Warm Winter" drive.
- **Cohort Representative:** MAE Department student council liaison.

References

Recommenders available upon request.

Education

Nazarbayev University — Expected June 2027

B.S. in Mechanical & Aerospace Engineering

GPA: 3.86 / 4.00 (Top 1 of Cohort) • Dean's List (4 semesters)

University of Hong Kong — Spring 2026

International Exchange Program

Collaborated with Prof. N. Fang, Prof. W. Yan & Prof. H. Kazerooni

Research & Experience

Research Assistant — TENG Devices — Jan – Dec 2025

Nazarbayev University

- Co-authored paper published in *Small, IF = 11.8, Q1*; designed and fabricated Triboelectric Nanogenerators (TENG) using electrospinning, and silicon printing.
- Improved device power density, endurance by **15–20%**; awarded Best Research Poster at the Belt & Road Initiative Conference 2025.

Product Support Engineering Intern — Jun – Jul 2025

Wabtec Corporation

- Conducted thermal testing and analysis of electric motors; implemented process changes that reduced material waste by **10%**; requested a product change to modify old technical drawings.

Research Assistant — Nuclear Reactor Design — Summer 2026

Nazarbayev University (Prof. Amgad Salama)

- Upcoming role focusing on computational core modeling and fluid-dynamics analysis for advanced nuclear reactor systems.

Projects

Ergonomic Train-Driver Seat Design

Mechaton Engineering Case, Sponsored by Wabtec Corporation, NU IMechE

- Analyzed structural loading on a technical chair for train drivers seated up to 12 hrs/day, benchmarking against **GOST 33330-2015** standards; verified the armrest, backrest, and seat against required force thresholds and developed a compliant CAD model with justified engineering calculations.

Transmission Gearbox Design — HKU Exchange

Group Project

- Designed and analyzed the input, intermediate, and output shafts of a multi-stage transmission gearbox: derived torque/speed at each stage from the gear ratios, sized shaft diameters against combined bending-torsional stress (ASME shaft equation), and selected keyways and bearings to meet a target factor of safety.

Savonius Wind Turbine & Turbidity Sensor

Group Project, Team Lead, Nazarbayev University

- Led a small team through design, simulation, and prototyping of a Savonius wind turbine, plus a turbidity sensor for a water filtration system.

Interdisciplinary Engineering Challenges — NU AIChE

Team Competition, NU AIChE Student Chapter

- **Avatar-themed:** Solved applied kinematics, dynamics, and thermodynamics problems framed as creative story-based scenarios.
- **Soccer-themed:** Tackled interdisciplinary chemistry, physics, and engineering problems with Prof. Dmitry Sizov (MAE Dept.), requiring rapid, strategic thinking.

3D-Modeled Sculptural Artwork — HKU Exchange

With Prof. H. Kazerooni (UC Berkeley)

- Designed and 3D-modeled and printed an artistic flower with petals blending CAD technique with creative form.